

Definition 1 (Planning task). A STRIPS planning task over a finite set of objects O is a tuple $P = \langle S, A, I, G \rangle$. Its atoms are the ground expressions $p(o_1, \dots, o_k)$ for a predicate p of arity k and $o_i \in O$. A state $s \in S$ is a set of atoms, $I \in S$ is the initial state, G is a set of atoms, and every action $a \in A$ is a triple $\langle \text{pre}(a), \text{add}(a), \text{del}(a) \rangle$ of sets of atoms.

Definition 2 (Applicability and progression). An action $a \in A$ is *applicable* in a state $s \in S$ if $\text{pre}(a) \subseteq s$, and the state it leads to is then

$$s[a] = (s \setminus \text{del}(a)) \cup \text{add}(a).$$

A plan is a finite sequence $\pi = a_1 \dots a_n$ of actions. The empty plan ε is applicable in every state, with $s[\varepsilon] = s$, and $\pi = a_1 \dots a_n$ with $n \geq 1$ is *applicable* in s if a_1 is applicable in s and $a_2 \dots a_n$ is applicable in $s[a_1]$. The plan then leads to $s[\pi] = s[a_1][a_2 \dots a_n]$. A plan π *solves* P if it is applicable in I and $G \subseteq I[\pi]$.

Definition 3 (Object collapse). Fix a set $C \subseteq O$ of objects to collapse, with $|C| > 1$, and a fresh symbol $c \notin O$. The collapse is the map $\sigma: O \rightarrow (O \setminus C) \cup \{c\}$ given by

$$\sigma(o) = \begin{cases} c & \text{if } o \in C, \\ o & \text{otherwise.} \end{cases}$$

Definition 4 (Abstraction map). The abstraction map $(\cdot)^*$ lifts σ to atoms, states, actions and plans by setting

$$\begin{aligned} p(o_1, \dots, o_k)^* &= p(\sigma(o_1), \dots, \sigma(o_k)), & s^* &= \{f^* : f \in s\}, \\ a^* &= \langle \text{pre}(a)^*, \text{add}(a)^*, \text{del}(a)^* \rangle, & (a_1 \dots a_n)^* &= a_1^* \dots a_n^*. \end{aligned}$$

Definition 5 (Abstract task). The abstraction of P under σ is $P^* = \langle S^*, A^*, I^*, G^* \rangle$, where S^* collects the states over $(O \setminus C) \cup \{c\}$, $A^* = \{a^* : a \in A\}$, and $(\cdot)^*$ carries I and G to I^* and G^* .

Definition 6 (Relaxed abstract task). Say that an atom $p(o_1, \dots, o_k)$ *mentions* an object o if $o \in \{o_1, \dots, o_k\}$. For $a \in A$ put

$$\text{del}^r(a) = \{d^* : d \in \text{del}(a) \text{ and } d \text{ mentions no object of } C\},$$

$$a^r = \langle \text{pre}(a)^*, \text{add}(a)^*, \text{del}^r(a) \rangle,$$

so that a^r forgets every delete whose atom mentions a collapsed object. The *relaxed abstract task* is $P^r = \langle S^*, A^r, I^*, G^* \rangle$ with $A^r = \{a^r : a \in A\}$. It differs from P^* only in the delete lists of its actions, and coincides with P^* when no delete of P mentions an object of C .

Since σ maps every object of C onto the single symbol c it is not injective, so $(\cdot)^*$ may identify distinct atoms, states and actions of P .

Lemma 1. *If $a \in A$ is applicable in $s \in S$, then a^r is applicable in s^* .*

Proof. By Definition 6 the relaxation leaves preconditions untouched, so $\text{pre}(a^r) = \text{pre}(a)^*$. From $\text{pre}(a) \subseteq s$ we get $\text{pre}(a)^* \subseteq s^*$, hence $\text{pre}(a^r) \subseteq s^*$. $\square$

Lemma 2 (Progression). *If $a \in A$ is applicable in $s \in S$, then $(s[a])^* \subseteq s^*[a^r]$.*

Proof. Progression gives $s[a] = (s \setminus \text{del}(a)) \cup \text{add}(a)$, so

$$(s[a])^* = (s \setminus \text{del}(a))^* \cup \text{add}(a)^*, \quad s^*[a^r] = (s^* \setminus \text{del}^r(a)) \cup \text{add}(a)^*,$$

and it suffices to show that $(s \setminus \text{del}(a))^* \subseteq s^* \setminus \text{del}^r(a)$. Let $f \in (s \setminus \text{del}(a))^*$. Then $f = g^*$ for some $g \in s$ such that $g \notin \text{del}(a)$, so $f \in s^*$. Suppose $f \in \text{del}^r(a)$. By Definition 6 we then have $f = d^*$ for some $d \in \text{del}(a)$ mentioning no object of C . Since σ fixes every object outside C , it fixes d , so $f = d$. Now d is an atom over O and $c \notin O$, so f does not mention c . If g mentioned an object of C then g^* would mention c , and $g^* = f$ does not. Hence g mentions no object of C either, so σ fixes g and $g = g^* = f = d$. This contradicts $g \notin \text{del}(a)$, so $f \notin \text{del}^r(a)$. $\square$

Lemma 3 (Monotonicity). *Let $s \subseteq t$ be states and let π be applicable in s . Then π is applicable in t and $s[\pi] \subseteq t[\pi]$.*

Proof. By induction on the length of π . The empty plan is applicable in every state and $s[\varepsilon] = s \subseteq t = t[\varepsilon]$. For $\pi = a\pi'$, applicability of a in s gives $\text{pre}(a) \subseteq s \subseteq t$, so a is applicable in t , and

$$s[a] = (s \setminus \text{del}(a)) \cup \text{add}(a) \subseteq (t \setminus \text{del}(a)) \cup \text{add}(a) = t[a].$$

The claim follows by applying the induction hypothesis to π' at $s[a] \subseteq t[a]$. $\square$

Lemma 4. *If π is applicable in $s \in S$, then π^r is applicable in s^* and $(s[\pi])^* \subseteq s^*[\pi^r]$.*

Proof. By induction on the length of π . For the empty plan both sides are s^* . Let $\pi = a\pi'$ with a applicable in s . By Lemma 1 a^r is applicable in s^* , and by Lemma 2 $(s[a])^* \subseteq s^*[a^r]$. The induction hypothesis applied to π' at $s[a]$ gives that π'^r is applicable in $(s[a])^*$ with $(s[a][\pi'])^* \subseteq (s[a])^*[\pi'^r]$. Since $(s[a])^* \subseteq s^*[a^r]$, Lemma 3 lets us move both to $s^*[a^r]$, where π'^r is applicable and

$$(s[\pi])^* = (s[a][\pi'])^* \subseteq (s[a])^*[\pi'^r] \subseteq s^*[a^r][\pi'^r] = s^*[\pi^r]. \quad \square$$

Theorem 1 (Over-approximation). *If $G \subseteq I[\pi]$, then $G^* \subseteq I^*[\pi^r]$.*

Proof. By Lemma 4 the plan π^r is applicable in I^* and $(I[\pi])^* \subseteq I^*[\pi^r]$. Applying $(\cdot)^*$ to $G \subseteq I[\pi]$ gives $G^* \subseteq (I[\pi])^*$, so

$$G^* \subseteq (I[\pi])^* \subseteq I^*[\pi^r]. \quad \square$$